Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_39

1. Summary of Key Deviations

The analysis of Run_39 reveals a set of minor, but consistent, physical deviations, and notably, the absence of any critical sensor failures.

- **p_Combustion (Minor):** The most consistent and primary anomaly is a minor **negative offset** on `p_Combustion`, which reads approximately **-5% to -6%** lower than healthy across all active states (s2–s7).
- **Flow_Bypass (Minor):** A corresponding minor positive **offset and drift** is observed, with flow increasing by up to **+8.6%**. This indicates more flow is being diverted than normal.
- **Flow_Pump (Minor):** A slight positive **offset** of around **+5%** is present, suggesting a minor over-delivery from the pump.
- **Flow_Engine1-4 (Minor):** All engine flows show a slight positive **offset** of **+3.5%**, but only during the high-throttle phase (s5).
- **Flow_PRV:** For the first time, this signal shows **no deviation**. This is a critical piece of evidence, confirming the Pressure Relief Valve and its sensor are operating nominally.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Combustion Chamber / Injector Nozzle(s)**	**Minor Leak**	The consistent **negative offset in p_Combustion** is the most likely a real physical symptom. A small leak in the combustion chamber or at one or more of the injector nozzles would cause a pressure drop. The fuel system would attempt to compensate by slightly increasing flow (**Flow_Pump** and **Flow_Engine** are both slightly positive), which is consistent with the observations.

2	**p_Combustion Sensor**	**Calibration Shift (Negative Offset)**	An alternative explanation is that the `p_Combustion` sensor has a minor negative calibration error. This would make it *appear* as if pressure is low. Under this assumption, the slightly increased flows could be a result of the control system reacting to the (erroneously) low-pressure reading, or they could be unrelated minor deviations.
3	**Fuel Metering Unit (FMU)**	**Slight Over-delivery / Calibration Shift**	The FMU could be delivering slightly more fuel than commanded, leading to the positive offset on **Flow_Engine1-4** in state s5. This could contribute to the other flow deviations but doesn't directly explain the primary symptom of low combustion pressure.
4	**Main Fuel Pump**	**Minor Over-performance**	The pump delivering ~5% extra flow is a possibility. This would explain the higher `Flow_Pump` and `Flow_Bypass` readings. However, this contradicts the primary symptom of *low* `p_Combustion`, as higher flow should ideally lead to higher pressure.
5	**Bypass Valve**	**Minor Calibration Shift (Opening too soon)**	The bypass valve could be opening at a slightly lower pressure than specified, causing a minor increase in bypass flow. This would be a consequence of another fault rather than the root cause itself.

3. Most Probable Hypothesis

The most probable root cause is a **Minor Leak in the Combustion Chamber or at an Injector Nozzle.**

Reasoning: This single physical fault provides the most coherent explanation for the entire set of symptoms. A leak downstream of the FMU would cause a direct loss of pressure in the combustion chamber (`p_Combustion` is low). The system's control logic would likely try to compensate for this pressure loss by increasing fuel flow, which is consistent with the minor positive offsets seen in `Flow_Pump`, `Flow_Bypass`, and `Flow_Engine`. Unlike a sensor fault, a real physical leak accounts for both the primary pressure drop and the secondary flow increases. The fact that `Flow_PRV` remains healthy rules out any issues with the high-pressure relief circuit, further isolating the problem to the combustion section.